

TASK 2 SUBMISSION

Competitor Demand Prediction Using Job Postings

Data Collection

Data Collection Report for NVIDIA

Prepared by	Nayab Khalid
Company specialism	NVIDIA
Task	Task 2 of 10 - Data Collection
Date of collection	2 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Result	Value
Distinct postings collected	1,745
Observation window	2026-05-13 to 2026-09-02, 12 weekly observations
Open roles per observation	395 to 433, mean 416
Postings opened / closed in window	1,579 / 1,561
Status at final observation	413 active, 1,332 closed
Job description text	0 of 1,745
Validator result	FAILED, 2 errors, both on the missing job_description field

Headline finding

Both sources approved in Task 1 were tested at the start of Task 2 and both failed. The Kaggle dataset is a four-week snapshot, not a 24-month history, and yields 11 NVIDIA postings. The cross-check dataset yields 271, covering data roles only. The source actually used was found during collection: an openly licensed daily snapshot of NVIDIA's Workday feed, which gives 1,745 postings and a genuine weekly time series, but no description text.

1 Source used

Field	Detail
Dataset	Aramente/eu-tech-jobs
URL	huggingface.co/datasets/Aramente/eu-tech-jobs
Licence	CC BY-4.0, attribution required
Origin of the NVIDIA rows	NVIDIA's public Workday careers feed, collected by the dataset maintainer
Company key	nvidia-workday
Coverage used	snapshots/<date>/jobs.parquet, sampled weekly
Accessed	2 September 2026

The dataset publishes a daily snapshot of open roles for a curated list of technology employers, with a companion companies table describing each one. NVIDIA is present as **nvidia-workday**, described there as a major Workday-hosted employer with EU offices.

2 Why the Task 1 sources were not used

Task 1 approved the Kaggle LinkedIn Job Postings dataset as the primary source and the Hugging Face data_jobs dataset as a cross-check. Both were tested at the start of Task 2 and both failed, for reasons that were not visible from their dataset descriptions. The reasoning is recorded here because it matters more than a tidy answer.

2.1 Kaggle, LinkedIn Job Postings 2023-2024

The description says the data covers 2023 and 2024, which I read in Task 1 as a 24-month history. It is not. It is a single snapshot of about 124,000 LinkedIn postings captured over roughly four weeks. Filtered to NVIDIA it yields **11 postings**. Eleven postings over four weeks cannot support hiring-trend analysis or forecasting.

This was tested by downloading an openly accessible mirror carrying the same schema plus a company_name column: 123,849 rows, 517 MB uncompressed. The Kaggle account token was therefore never the obstacle. The data itself is too thin for NVIDIA.

2.2 Hugging Face, lukebarousse/data_jobs

785,741 rows, Apache-2.0, downloaded and inspected in full. Filtered to NVIDIA it yields **271 postings** across 12 months of 2023. That is a usable monthly series, but it carries no description text and covers only data-family roles, so it represents a slice of NVIDIA's hiring rather than NVIDIA's hiring.

It also proved unusable as the shared cross-company basis proposed in the Task 1 collection plan: Anthropic appears in it 9 times, against 752 for Google and 646 for Microsoft.

A data-quality finding worth carrying to the whole team

NVIDIA appears in that dataset under 11 different employer strings, including NVIDIA, Nvidia, NVIDIA Corporation, and entity codes such as CN05 NVIDIA Shanghai WFOE and IN02 NVIDIA GraphicsPLtd,Pune. Any member filtering on an exact company name will silently lose rows. Every member should apply a name-matching rule rather than an exact match.

2.3 Other sources tested

edwarddgao/open-apply-jobs, MIT licensed, publishes daily parquet files from Greenhouse, Ashby and Lever with full description text. It contains no NVIDIA data at all, because NVIDIA publishes through Workday and that dataset covers only those three applicant tracking systems. It does carry Anthropic (431 roles), OpenAI (644), Databricks (855) and Snowflake (410), so it is worth raising with the members whose companies use those platforms.

3 Legal checks

Check	Result
Licence of the dataset used	CC BY-4.0, confirmed 2026-09-02 from the dataset card metadata
Attribution requirement	Recorded in the source column of every row, and cited in this report
Automated access to NVIDIA's own site	Not performed. NVIDIA's Terms of Service section 3.2 prohibits it, as established in Task 1
Personal data	None collected. No field in the source carries applicant or recruiter information
Validator PII scan	Clean. No email addresses or telephone numbers detected
Bypassing technical controls	None. All files downloaded from the dataset's public HTTPS endpoints

Provenance, stated plainly

The NVIDIA rows originate from NVIDIA's Workday feed, which NVIDIA's own Terms of Service prohibit *me* from collecting automatically. I did not collect it. A third party did, and published the result under CC BY-4.0. Under the rule the team agreed in Task 1, section 7 of the legal rules, this is third-party derived data: used as a documented secondary source, labelled as such in every output, and never presented as a first-party collection. This is the same position the team took on the LinkedIn-derived datasets.

4 Method

The collection is reproducible by running `collect_nvidia.py` from the repository. Three decisions in it are worth stating.

4.1 Snapshots, not diffs

The dataset also ships daily diff files. They are not used. Across the whole dataset the diff stream carries new, removed and changed events, but for NVIDIA it carries only new: **8,797 new events for 2,029 distinct postings**, the same posting re-announced up to 18 times and never once marked removed. Flow figures derived from it would be fiction. The snapshots are internally consistent, so first-seen and last-seen are derived from them instead.

4.2 Weekly sampling

126 daily snapshots are available. Sampling every seventh day gives 19 candidate observations and keeps the download near 350 MB rather than 2.4 GB. Weekly ISO buckets are already the team standard in the shared analysis configuration.

4.3 Partial scrapes excluded

7 of the 19 sampled observations returned far fewer NVIDIA rows than the rest. These are failed collections on the source's side, not real collapses in hiring, and were excluded. The rule applied is stated in the script: exclude any observation below 60 per cent of the median, which is about 416.

Excluded observation	NVIDIA rows returned
2026-04-29	0
2026-05-27	9
2026-06-03	9
2026-05-06	57
2026-07-22	66
2026-07-15	168
2026-06-17	171

Job identifiers were verified as stable before deriving flows: between the 2026-08-27 and 2026-09-02 snapshots, all 311 postings whose URL appears in both carry the same identifier in both. Appearances and disappearances therefore reflect real openings and closures rather than churning keys.

5 Files produced

File	Rows	Size	Contents
data/raw/nvidia_postings_raw_20260902.csv	1,745	753 KB	The dataset. One row per distinct posting, 25 columns, in the team schema
collect_nvidia.py	-	-	The collection script, re-runnable from a clean checkout

There is deliberately **one** dataset file. An earlier draft also shipped a separate weekly series, but that was an aggregate of this file at a different grain, and shipping two files invited the question of which one is the submission. The weekly series in section 5.2 is generated on demand by the script and is exactly reconstructable from the observed_dates column, verified observation by observation across all 12 weeks.

The naive reconstruction does not work, and the column exists because of it. Postings disappear and reappear between observations, so treating first_seen to last_seen as a continuous open span overcounts the stock: it gives 499 open roles on 2026-06-10 against a true 430. Storing the exact observation dates on which each posting was present makes the series recoverable without error.

5.1 Field coverage

Field	Populated	Note
job_id, company_name, job_title, job_url	1,745 / 1,745	Complete
posting_date	1,745 / 1,745	Proxy, see limitations
location	1,745 / 1,745	But 96 per cent are collapsed, see limitations
job_category	878 / 1,745	engineering 655, ml-ai 87, sales 35, research 26, ops 23
seniority_level	662 / 1,745	senior 573, principal 32, junior 19, mid 18, exec 8
first_seen, last_seen, weeks_observed, job_status	1,745 / 1,745	Derived by the script
observed_dates	1,745 / 1,745	JSON list of the exact observation dates the posting was present
job_description	0 / 1,745	The source carries none for NVIDIA

Field	Populated	Note
tech_stack_tags, countries, salary, employment_type	0 / 1,745	Empty in the source for NVIDIA

5.2 Weekly series

Observation	ISO week	Open roles	New	Closed	Net
2026-05-13	2026-W20	395	-	-	-
2026-05-20	2026-W21	411	109	93	+16
2026-06-10	2026-W24	430	266	247	+19
2026-06-24	2026-W26	417	132	145	-13
2026-07-01	2026-W27	403	88	102	-14
2026-07-08	2026-W28	415	97	85	+12
2026-07-29	2026-W31	419	297	293	+4
2026-08-05	2026-W32	421	105	103	+2
2026-08-12	2026-W33	419	115	117	-2
2026-08-20	2026-W34	419	122	122	0
2026-08-27	2026-W35	433	146	132	+14
2026-09-02	2026-W36	413	102	122	-20

Gaps between observation dates are uneven because excluded weeks were dropped. The new and closed columns count change since the previous *retained* observation, so the three intervals spanning a dropped week cover more than seven days and are not directly comparable with the others. This must be normalised per day before the series is used in Task 5.

6 Validation

```
python scripts/validate_dataset.py \
  work/task-2/nvidia-nayab-khalid/data/raw/nvidia_postings_raw_20260902.csv

rows          1745
unique job_ids 1745
company_name   NVIDIA
date range     2026-05-13 to 2026-09-02 (5 months)

ERROR 1745 rows have an empty required field: job_description
ERROR job_description shorter than 20 chars: row 1: 0 chars; ...

FAILED - 2 error(s), 0 warning(s).
```

The dataset fails the shared schema on one required field, `job_description`, because the source does not carry it for NVIDIA. Per the schema rule the column is retained and left empty rather than dropped. Everything else passes: no duplicate identifiers, no empty identifiers, all dates ISO-8601, one company per file, no personal data.

This needs a team decision, not a quiet edit

Either the required-field rule is relaxed for sources that cannot supply descriptions, or members without description text run a reduced version of Tasks 3 and 4. It should not be resolved by editing my file until it passes.

6.1 A fix made to the shared validator

The validator's phone-number heuristic matched the bare string `te1`, so every NVIDIA job URL containing Israel-Tel-Aviv was reported as a suspected telephone number. The pattern now requires word boundaries, and URL and identifier columns are excluded from the phone scan. Verified in both directions afterwards: a fixture containing a real email address and a real telephone number is still caught, and the NVIDIA URLs no longer produce false positives.

7 Limitations

These carry forward into every later task and must appear in the Task 9 report.

#	Limitation	Consequence
1	No job description text. 0 of 1,745 postings.	Skill extraction in Task 4 must work from job titles, which contain far fewer skills than a description. Comparisons against members who do have description text are not like-for-like.
2	<code>posting_date</code> is a proxy. Workday exposes no posting date, so the first weekly observation is used.	Postings already open on 2026-05-13 are left-censored. Duration and time-to-fill analysis is not supported.
3	Weekly resolution.	A posting opened and closed within one week is never observed, so turnover is understated by an unknown amount.
4	Location is mostly collapsed: 1,674 of 1,745 read as "2 Locations" or similar, only 71 are named.	Regional analysis is effectively unavailable for NVIDIA.
5	European bias. The source curates employers with EU offices and remote-EU hiring.	This is NVIDIA's European hiring, not global. Comparison with a member using a global source will overstate the difference in scale.
6	Under four months of coverage, 12 observations.	Seasonality cannot be estimated, and a 12-week forecast horizon fitted on 12 observations is thin. Task 7 must report wide intervals and lean on a naive baseline.
7	Third-party derived, completeness against NVIDIA's actual feed unverified.	Stated in every output that uses it.

8 Recommendations for the team

- Ratify the description question in the sprint meeting.** Four members will not all have description text. Either Tasks 3 and 4 run on titles for everyone, so the comparison is fair, or the difference is documented and accepted.
- Reconsider the shared comparison basis.** The Task 1 plan proposed that all four members pull their company's rows from one common dataset. No dataset tested holds all four companies at usable volume: `open-apply-jobs` covers Anthropic well and NVIDIA not at all, `data_jobs` covers Google and Microsoft well and Anthropic barely.
- Canonicalise company names on load.** NVIDIA appears under 11 distinct employer strings in one source tested. Exact-match filtering silently loses rows.

4. **Keep collecting weekly.** The snapshot source updates daily. Re-running the script each week extends the series and directly improves Task 7, which is currently short of observations.

Attribution

Job posting data for NVIDIA is taken from the Aramante/eu-tech-jobs dataset, published on Hugging Face under the Creative Commons Attribution 4.0 International licence. Datasets tested and not used are credited in section 2: arshkon/linkedin-job-postings (CC BY-SA 4.0) via the MichaelYitzchak/LinkedInJobPostings mirror, lukebarousse/data_jobs (Apache-2.0), and edwarddgao/open-apply-jobs (MIT).

Declaration	No personal or sensitive data has been collected, stored or committed. No automated request was made to any NVIDIA-operated site. All checks recorded here were performed on 2 September 2026 and are reproducible by running the collection script.
Prepared by	Nayab Khalid, NVIDIA specialist
Date	2 September 2026